

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Circles, Arcs & Sectors

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

CIRCLES, ARCS & SECTORS · CH4 · L6



ENG. ESLAM AHMED

The Map

This lesson collapses Circles, Arcs & Sectors into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Find sector area

TRIGGER *"work out", "find", a missing value*

03 Find arc length

TRIGGER *"work out", "find", a missing value*

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 41 website questions, 41 checked solutions, 4 local PDFs read.

01 Use fraction of the circle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1H Q8

The inner radius is m

This is a reasoning step: write one clear sentence, then continue the calculation.

Calculate value

$$7 + 2 = 9 \text{ m}$$

Finish: 54.3 m.

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

02 Find sector area

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$7 + 2 = 9 \text{ m}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
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— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

8 The region, shown shaded in the diagram, is a path. Diagram NOT accurately drawn
2 m O 7 m The boundary of the path is formed by two semicircles, with the same centre O, and two straight lines. The inner semicircle has a radius of 7 metres. The path has a width of 2 metres. Work out the perimeter of the path. Give your answer correct to one decimal place. m

YOUR SOLUTION

03 Find arc length

TRIGGER *"work out", "find", a missing value*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$5 = \frac{55}{360} \times 2\pi r$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2022 P1H Q16

Use trigonometry

$$5 = \frac{55}{360} \times 2\pi r$$

Area of the circle

$$\pi r^2 = \pi(5.208\dots)^2 = 85.233\dots$$

Finish: 85.2 cm².

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Combine sector with triangle/rectangle

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x . Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

05 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Circles, Arcs & Sectors question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$6 + x$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q22

The radius of sector is

$$6 + x$$

Use trigonometry

$$\frac{50}{360} \times 2\pi(6 + x)$$

Finish: 5.89

FORMULA BANK

$$\text{Arc} = \frac{\theta}{360} 2\pi r. \text{ Sector area} = \frac{\theta}{360} \pi r^2.$$

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

22 The diagram shows a sector OBC of a circle with centre O and radius $(6 + x)$ cm. A 50 degrees D x cm O B C Diagram NOT accurately drawn 6 cm A is the point on OB and D is the point on OC such that $OA = OD = 6$ cm Angle BOC = 50 degrees Given that the perimeter of sector OBC = 2 times the perimeter of triangle OAD find the value of x . Give your answer correct to 3 significant figures. $x =$

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use fraction of the circle
"work out", "find", a missing value	Find sector area
"work out", "find", a missing value	Find arc length
"work out", "show", "hence", a familiar exam phrase	Combine sector with triangle/rectangle
"work out", "show", "hence", a familiar exam phrase	Use trigonometry
FINAL BOTTOM LINE	Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.